

Policy Analysis - Estimating Treatment Effects

February 24, 2026

- What criteria should we use to assess existing evidence?
 - ▶ Strength of the methodology
 - Does the evidence provide a causal interpretation?
 - Is the evidence rooted in the appropriate historical context?
 - ▶ Quality of the data
 - ▶ External validity

- **Steps for conducting a quantitative policy evaluation.**

1. Develop a research question (hypothesis).
 - 1a. Assess the existing evidence.
 - 1b. What is your contribution?
2. Develop a research strategy.
 - 2a. Empirical methodology.
 - 2b. Cost effectiveness/benefit/utility analysis.
3. Identify appropriate data.
 - 3a. Primary vs. secondary data.
 - 3b. Cross-sectional vs. panel (longitudinal) data.
 - 3c. Power analysis?
4. Engage funders/stakeholders.
5. Construct an analytic sample (i.e., data management).
 - 5a. STATA, SAS, R, SQL, Excel, etc.
6. Conduct data analysis.
7. Report findings.

Policy Evaluation - Estimating Treatment Effects

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- Average Treatment Effect (ATE) of an intervention among a sample of n people:

$$ATE = Avg_n[Y_i^1 - Y_i^0]$$

Where Y_i^1 is the *potential* outcome for person i with treatment and Y_i^0 is the *potential* outcome for person i without treatment.

Estimating Treatment Effects

- Suppose we're interested in the effect of a new surgical intervention ($D_i = 1$) for cancer on longevity compared to standard chemotherapy ($D_i = 0$).

Outcomes for ten patients receiving surgery (Y^1) or chemotherapy (Y^0)

Patient	Y^1	Y^0
1	7	1
2	5	6
3	5	1
4	7	8
5	4	2
6	10	1
7	1	10
8	5	6
9	3	7
10	9	8

- Calculate the ATE ($ATE = Avg_n[Y_i^1 - Y_i^0]$).

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- Problem: We can never obtain the ATE this way because we don't observe the ***counterfactual***.

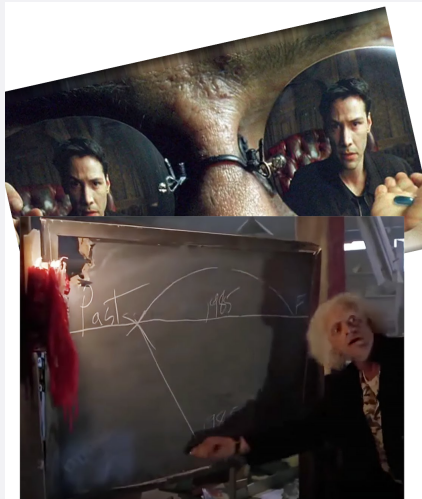
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- Estimated Effect of treatment on Y :

$$ATE_{est} = Avg_n[Y_i^1 | D_i = 1] - Avg_n[Y_i^0 | D_i = 0]$$

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6	10	1	1	10
7	1	10	0	10
8	5	6	0	6
9	3	7	0	7
10	9	8	1	9

- $ATE = ?$
- $ATE_{est} = Avg_n[Y_i^1 | D_i = 1] - Avg_n[Y_i^0 | D_i = 0] = ?$

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- $ATE_{est} = Avg_n[Y_i^1 | D_i = 1] - Avg_n[Y_i^0 | D_i = 0]$
- When does $ATE_{est} = ATE$ and when does it not?

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- If independence is violated, then treatment effects differ by treatment status:
 - ▶ Average Treatment Effect for the Treated:

$$ATT = Avg_n[Y_i^1 | D_i = 1] - Avg_n[Y_i^0 | D_i = 1]$$

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 - ▶ Average Treatment Effect for the Treated:

$$ATT = Avg_n[Y_i^1 | D_i = 1] - Avg_n[Y_i^0 | D_i = 1]$$

- ▶ Average Treatment Effect for the Untreated:

$$ATU = Avg_n[Y_i^1 | D_i = 0] - Avg_n[Y_i^0 | D_i = 0]$$

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- $ATE = ?$
- $ATE_{est} = Avg_n[Y_i^1 | D_i = 1] - Avg_n[Y_i^0 | D_i = 0] = ?$
- $ATT = Avg_n[Y_i^1 | D_i = 1] - Avg_n[Y_i^0 | D_i = 1] = ?$
- $ATU = Avg_n[Y_i^1 | D_i = 0] - Avg_n[Y_i^0 | D_i = 0] = ?$

- With no independence:

$$ATE = \pi ATT + (1 - \pi)ATU$$

where π is the share of the sample receiving treatment.

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- $ATE_{est} = ?$
- $ATT = ?$
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- $ATE = ATT(\pi) + ATU(1 - \pi) = ?$

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- Do some algebra:

$$\begin{aligned} ATE_{est} = ATE &+ \{Avg_n[Y_i^0 | D_i = 1] - Avg_n[Y_i^0 | D_i = 0]\} \\ &+ (1 - \pi)(ATT - ATU) \end{aligned}$$

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- With no independence:

$$ATE = \pi ATT + (1 - \pi) ATU$$

- Do some algebra:

$$\begin{aligned} ATE_{est} = ATE + & \underbrace{\{Avg_n[Y_i^0 | D_i = 1] - Avg_n[Y_i^0 | D_i = 0]\}}_{\text{Selection Bias}} \\ & + \underbrace{(1 - \pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias}} \end{aligned}$$